Advanced Computer Graphics (ACG)

Theory Session 4 – Rendering Equation, Monte Carlo Integration and Direct Illumination

2026-2027

Pompeu Fabra University

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Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Class Outline

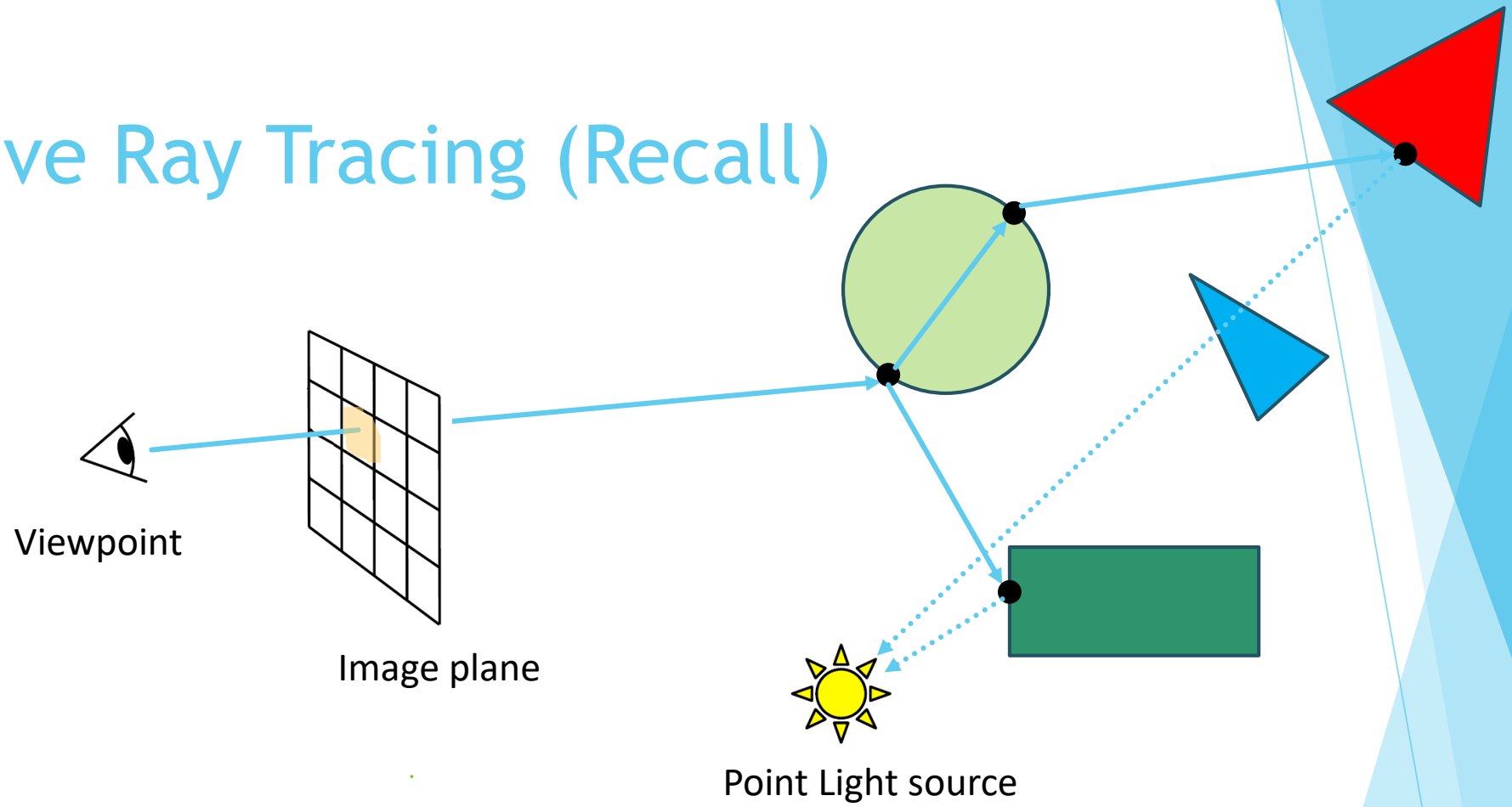
- Part I - Beyond RRT
  - Motivation
  - Basic Radiometry
  - The Rendering Equation
  - The Modified Phong Reflection Model
- Part II – Solving the Rendering Equation
  - Monte Carlo Integration
  - Direct Illumination
  - Area Formulation of the Rendering Equation



Part I
Beyond RRT

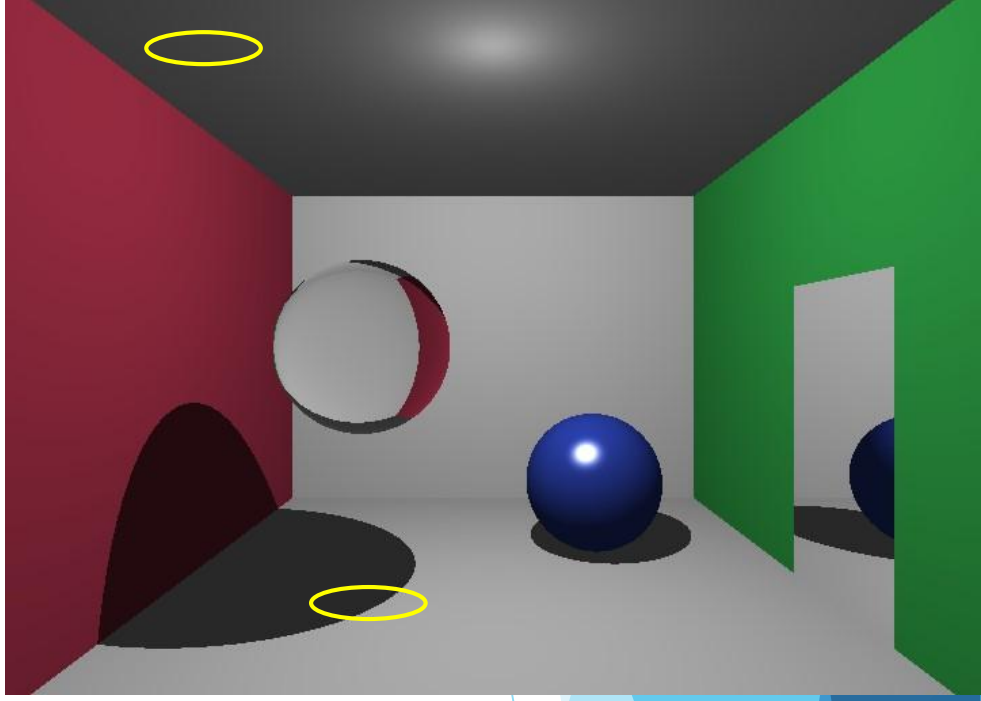
From Empirical to Physically-Based

Recursive Ray Tracing (Recall)



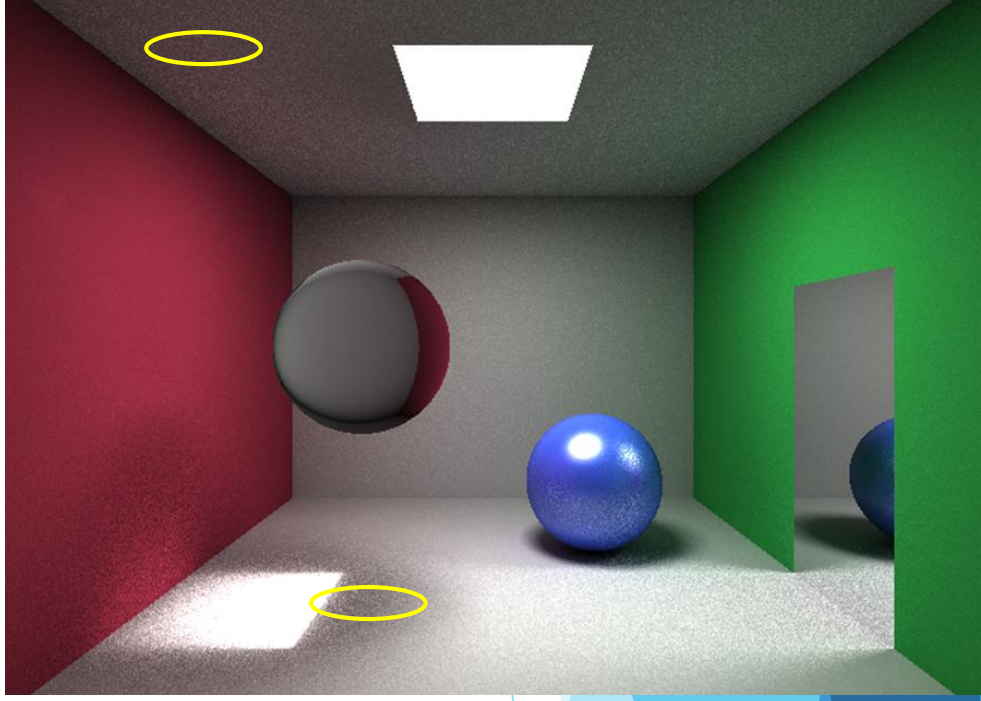
- Recursively trace secondary rays until a non-specular surface is hit
- At each intersection point, evaluate light sources visibility (no contribution if occluded)
- Final pixel color results from all light contributions along the paths
- Allows simulating specular reflections and transmissions

Motivation: what we have so far (Recursive Ray Tracing)



- Point light sources
  - Do not exist in reality ➔ Lack of realism
- Phong reflection model (diffuse and glossy reflections)
  - Totally empiric model (based on observations, no underlying theory)
- Local (i.e., direct) Illumination only
  - Only direct lighting is explicitly computed at each non-specular object
  - Global (indirect) illumination “guessed” using a per-scene constant (ambient term)
  - No caustics

Motivation: what we would like to have



- Our goal
  - Synthesize more realistic images
  - How?
  - Global Illumination: physically-based light propagation
- OK… but…
  - How do we do physically-based light propagation?
  - How do we measure light?
- **Today's class**: understanding of what is light and of the physical quantities that represent light energy

What is Light?

Light is electromagnetic radiation

- Waves of the electromagnetic field propagating through space
- Visible light has a wavelength λ in the range $\lambda \in [380, 780]$ nm
- Visible light propagates ***photons***, and can be analyzed both as a wave and as a **particle**

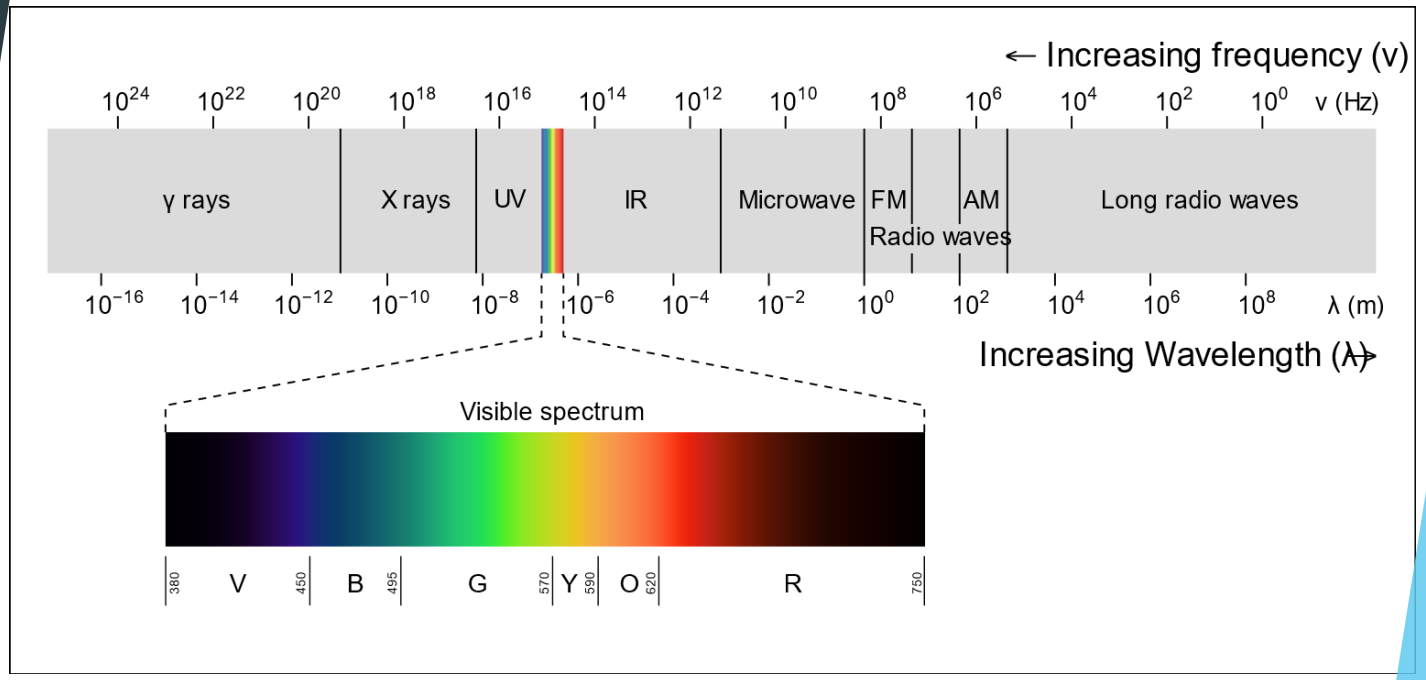
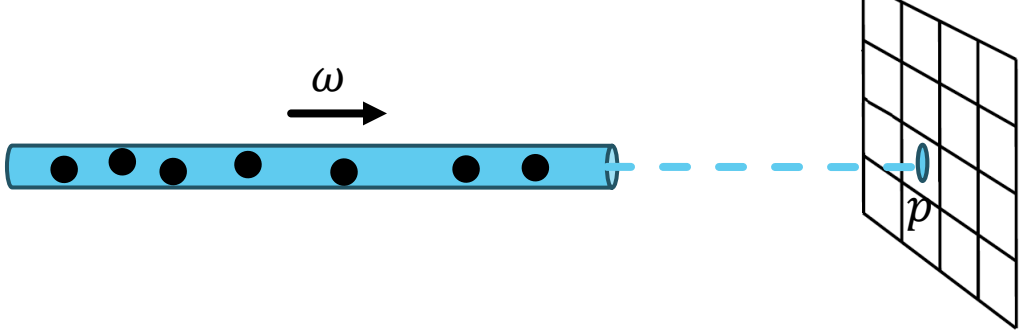


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https://commons.wikimedia.org/wiki/File:EM_spectrum.svg#/media/File:EM_spectrum.svg

Intuition



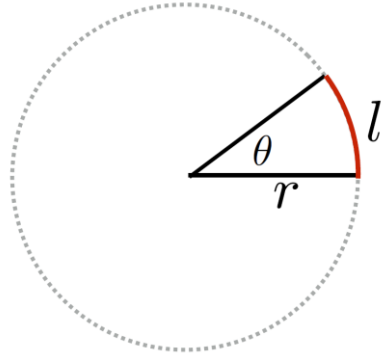
- Consider a **light beam** with direction ω propagating several photons which hit a small point p located in our image plane

- Informally, the "color" of the light hitting p from direction ω is determined by:
  - How many photons hit point p during a very small time interval $\mathrm{d}t$
  - The wavelength of the photons that hit p

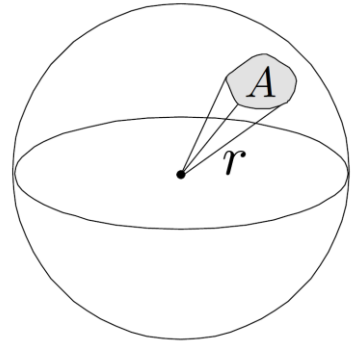
- The **physical quantity** which captures this phenomena is called ***Radiance***
  - Radiance captures the "appearance" of objects in a scene
  - Quantity associated with rays in a ray-tracer
  - Rendering is in fact all about computing radiance

Angles and Solid Angles

- An angle θ is defined as the ratio between the subtended arclength on the circle (l) and the circle radius (r)
  - $\theta = \frac{l}{r}$ (radians)
  - The circle subtends a solid angle of 2π radians

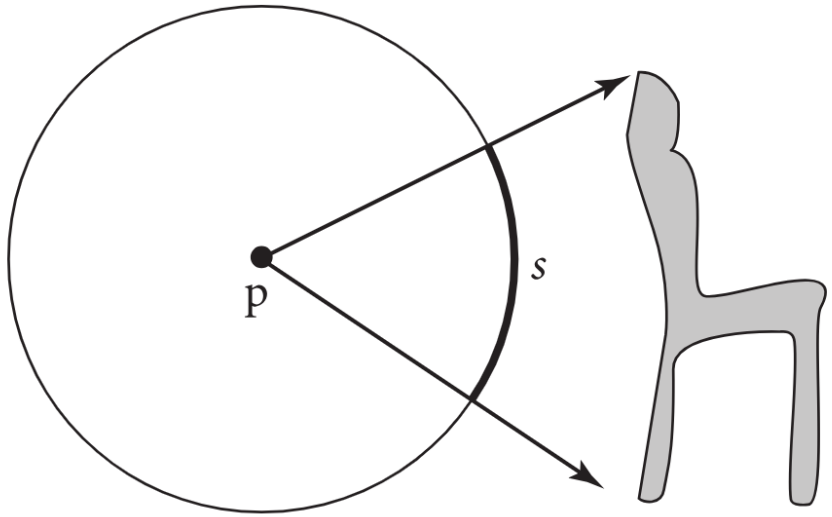


- A solid angle results from extending the notion of angles from 2D to 3D
- A solid angle Ω is defined as the ratio between the subtended area on the sphere (A) and the square of the radius (r^2)
  - $\Omega = \frac{A}{r^2}$
  - A 3D sphere subtends a solid angle of 4π steradians

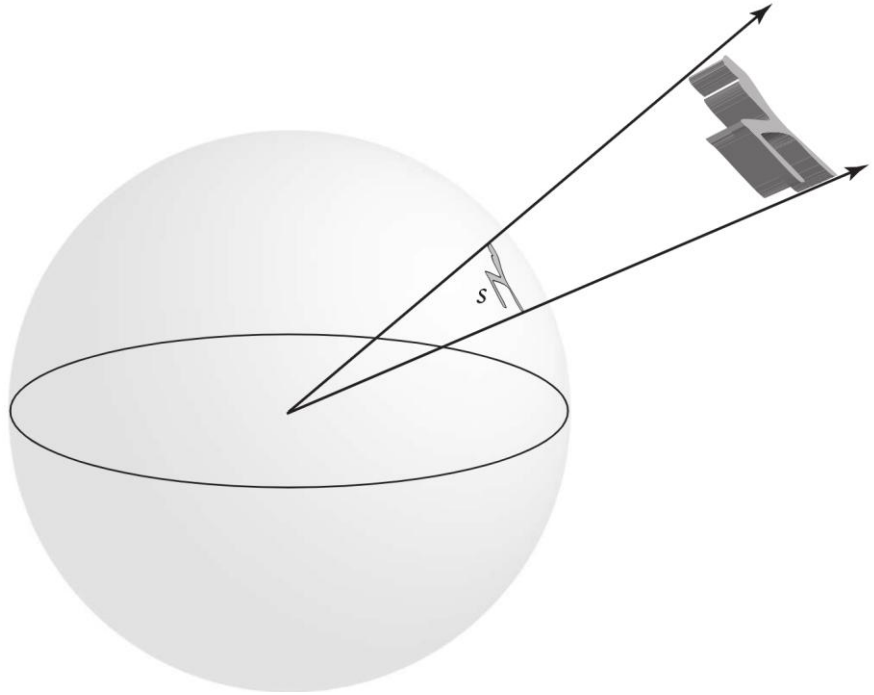


Example of a Solid Angle

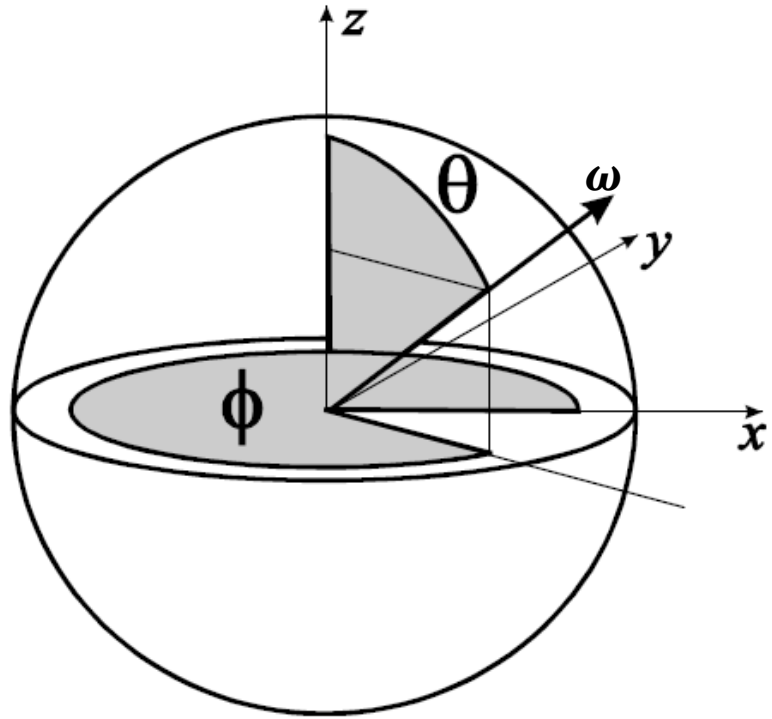
Subtended Angle (2D)



Subtended Solid Angle (3D)

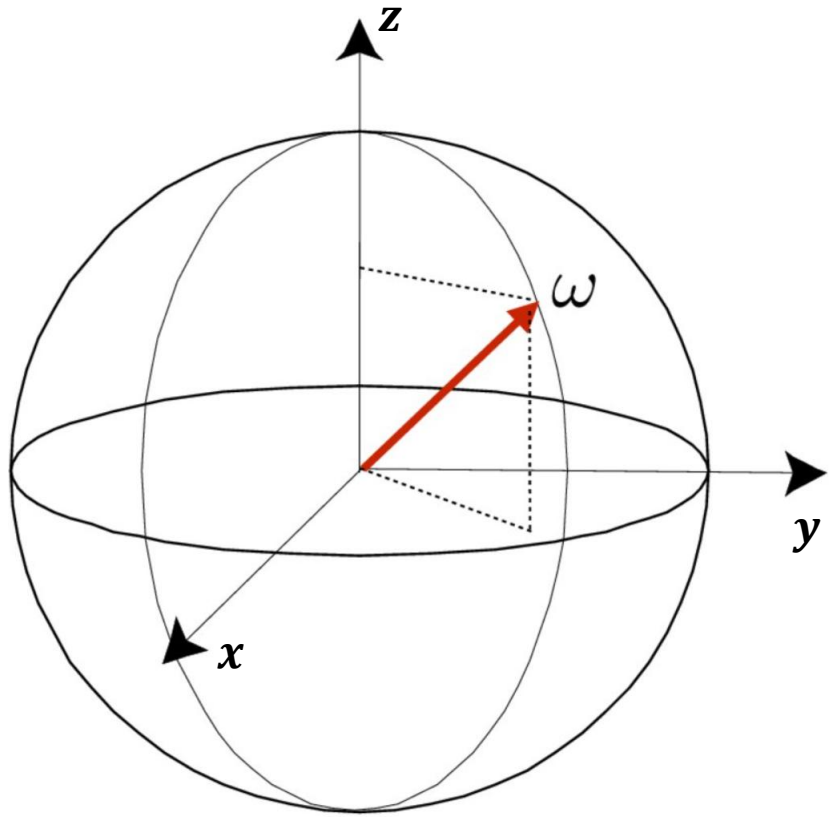


Spherical Directions



- To express a direction on the hemisphere, we usually use the small case Greek letter omega (ω)
- Spherical directions can be written in terms of spherical coordinates $\boldsymbol{\omega} = (\boldsymbol{\theta}, \boldsymbol{\phi})$, where:
  - $\theta \in [0, \pi]$ is the polar (or elevation) angle, measured with the z-axis
  - $\phi \in [0, 2\pi]$ is the azimuth (or rotation) angle, measured with the x-axis
- Spherical directions can also be expressed in terms of Cartesian coordinates, such that:
  - $\boldsymbol{\omega} = (\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$, with $\|\omega\| = 1$

From Spherical to Cartesian Coordinates (and Vice-Versa)



- It is easy to convert a spherical direction $\omega = (\theta, \phi)$ from spherical coordinates to Cartesian coordinates using:

$$x = \sin\theta \cos\phi$$
$$y = \sin\theta \ \sin\phi$$
$$z = \cos\theta$$

- Similarly, a spherical direction $\omega = (x, y, z)$ can be converted to spherical coordinates using:

$$\theta = \cos^{-1} z$$

$$\phi = \tan^{-1} \frac{y}{x}$$

Differential Solid Angle

- "Infinitesimally small" solid angle around a given direction
- By convention, represented as a 3D vector
  - Magnitude: $\mathrm{d}\omega$
    - Size of a differential area on the unit sphere
  - Direction: ω
    - Center of the projection of the differential area onto the unit sphere

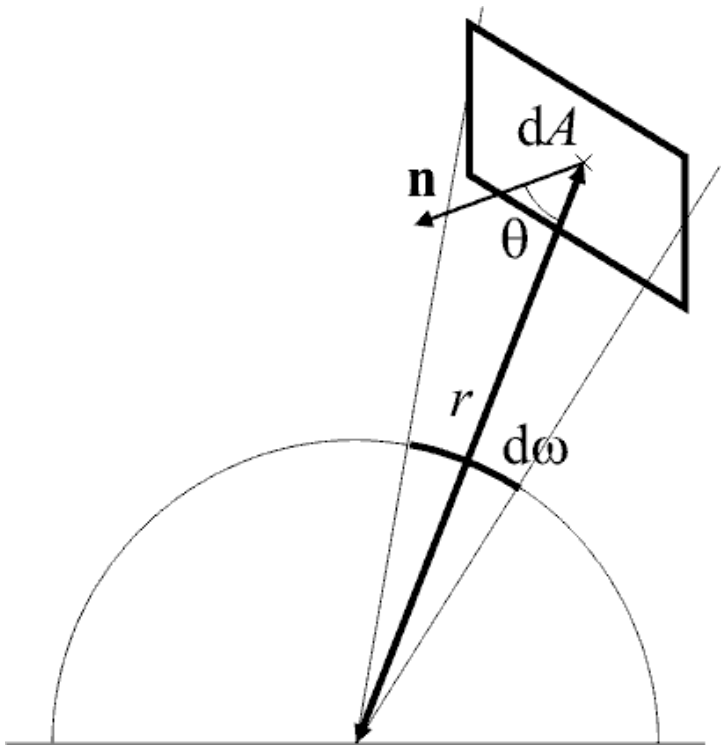
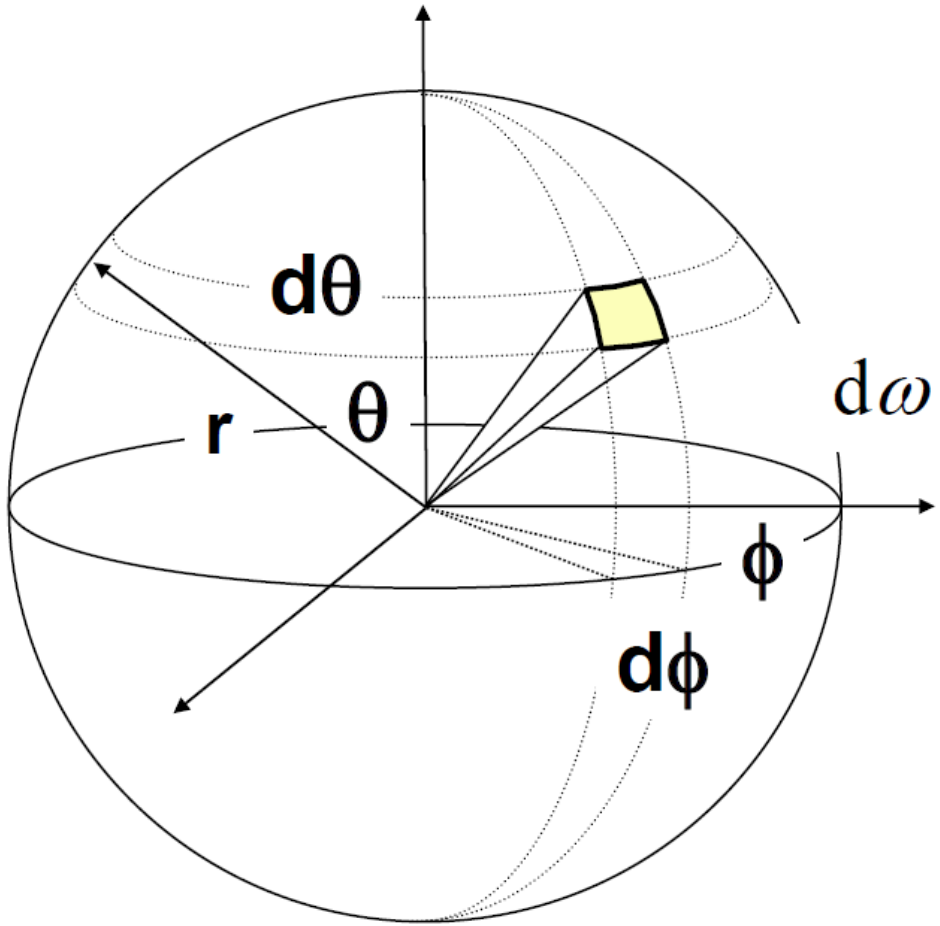


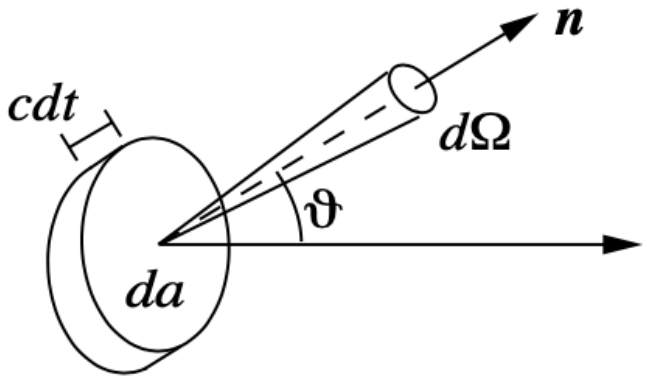
Image and text from:
CG III (NPGR010) - J. Křivánek



$$d\omega = (d\theta)\,(\sin\theta d\phi)$$
$$= \sin\theta d\theta d\phi$$

Differential Solid Angle

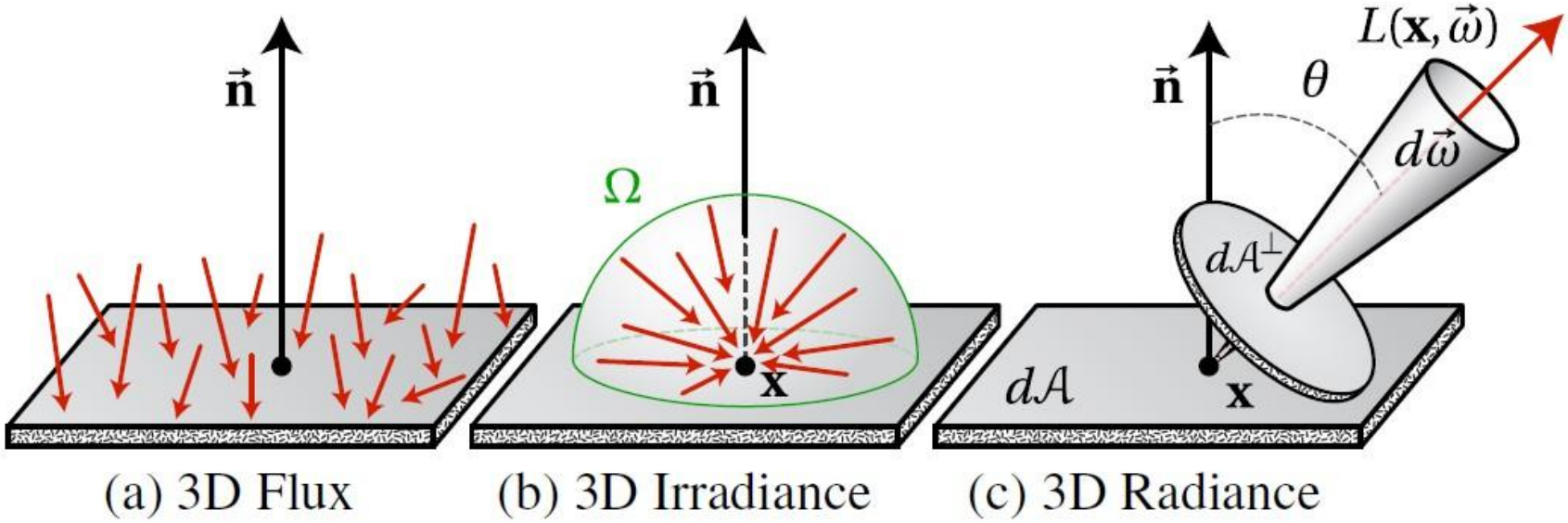
Radiance (and other Magnitudes)



- Radiance is one of a set of radiometric quantities used to describe the interaction of visible electromagnetic radiation (i.e., light) with matter

Radiometric Magnitude	Symbol	Units
Flux	Φ	W (watts=J/s)
Irradiance	E	W/m^2
Radiant Intensity	I	W/sr
Radiance	$\boldsymbol{L}$	**$W/(m^2\,sr)$**

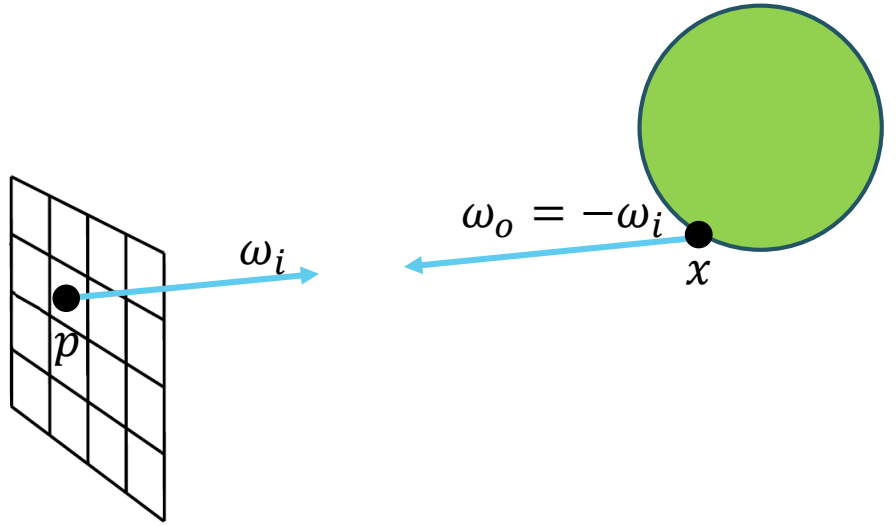
Summary of Basic Radiometric Quantities



The red arrows represent photon beams flowing through the 3D space

Image: Wojciech Jarosz

Synthesizing an image



- For each pixel p_i, compute the incident (spectral) radiance

- The incident radiance (L_i) at p from ω_i is the outgoing radiance (L_o) at x towards $\omega_o = -\omega_i$

$$L_i(\lambda, p, \omega_i) = L_o(\lambda, x, \omega_o)$$

where:

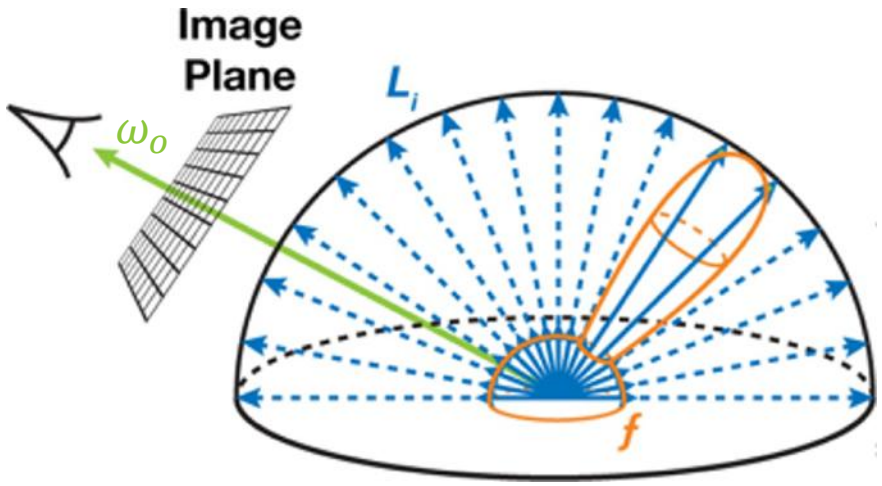
- $\Psi(p, \omega_i)$ is the ray casting operation
- Returns the closest visible point x along a ray originating at point $p$and pointing in the direction ω_i

$$L_o(\lambda, x, \omega_o) = ?$$

The Rendering Equation (RE)
Hemispherical Formulation

- The value of $L_o(\lambda, x, \omega_o)$ is given by the RE, which mathematically formulates the equilibrium distribution of light energy in a scene

$$L_o(\lambda, x, \omega_o) = L_e(\lambda, x, \omega_o) + \int_{\Omega_{2\pi}} L_i(\lambda, x, \omega_i)\, f_r(\lambda, \omega_o, \omega_i)\, (N \cdot \omega_i)\, \mathrm{d}\omega_i$$



Note: In the following, I will sometimes drop the wavelength term λ to ease notation

The Rendering Equation (RE)
Hemispherical Formulation

- The RE reads as follows (ignoring the wavelength argument)
  - The outgoing radiance L_o from point x towards direction ω_o is equal to the self-emitted radiance L_e from x towards ω_o, plus the sum (integral), over all possible incident light directions ω_i on the hemisphere $\Omega_{2\pi}$ centered on the normal at x, of the incident radiance at x from direction ω_i, multiplied by the material reflectance f for the pair of directions ω_i and ω_o, weighted by the cosine of the angle between N (the normal at x) and ω_i

$$L_o(x, \omega_o) = L_e(x, \omega_o) + \int_{\Omega_{2\pi}} L_i(x, \omega_i)\, f_r(\omega_o, \omega_i)\, (N \cdot \omega_i)\, \mathrm{d}\omega_i$$

Video 1 – Short (3:28)

Video 2 – A bit longer (9:23)

BRDF

$f_r(\omega_o, \omega_i)$

- Bi-directional Reflectance Distribution Function (BRDF)
  - Models the reflectance of light at the surface (i.e., the surface material properties)
  - Its arguments are two directions ω_o and ω_i
  - The result is invariant to swapping both directions (i.e., $f_r(\omega_o, \omega_i) = f_r(\omega_i, \omega_o)$)
  - Yields the fraction of incident light that is reflected at the surface for a given pair of directions

- Must respect the energy conservation law:

$$\forall \omega_o: \int_{\Omega_{2\pi}} f_r(\omega_o, \omega_i)\,(N \cdot \omega_i)\,\mathrm{d}\omega_i \leq 1$$

Phong Reflection Model

$$\forall \omega_o: \int_{\Omega_{2\pi}} f_r(\omega_o, \omega_i)\,(N \cdot \omega_i)\,\mathrm{d}\omega_i \leq 1$$

- Recall the Phong reflection model seen in the last class

$$f_r(\omega_i, \omega_o) = k_d + k_s(\omega_r \cdot \omega_o)^s$$

- Taking $\omega_o = N$ (i.e., supposing we are looking at the surface from above) yields:

$$\int_{\Omega_{2\pi}} f_r(\omega_i, N)\,(N \cdot \omega_i)\,\mathrm{d}\omega_i = k_d\pi + k_s\frac{2\pi}{s+2}$$

- If $k_d = k_s = 1$ then $k_d\pi + k_s\frac{2\pi}{s+2} > 1$ (violates the energy conservation law)

Modified Phong Reflectance Model

- By normalizing k_d and k_s, we can obtain a physically plausible Phong BRDF:

$$f_r(\omega_i, \omega_o) = \frac{k_d}{\pi} + \frac{s+2}{2\pi} k_s (\omega_r \cdot \omega_o)^s$$

- This formulation respects the energy conservation **as law as long as** $k_d + k_s \leq 1$

- Further Reading:
  - Using the Modified Phong Reflectance Model for Physically Based Rendering
  - Global Illumination Compendium (page 32)

Exercise 1

Knowing that the integral of a differential solid angle over the hemisphere can be expressed in terms of spherical coordinates as:

$$\int_{\Omega_{2\pi}} \mathrm{d}\omega = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

Show that, for $N = z = (0,0,1)$ and $\omega_o = N \rightarrow \omega_r = \omega_o = N$, the modified Phong reflectance model respects the energy conservation law.

Recall (some simple derivatives):

$$(\cos x)' = -\sin x$$
$$(\sin x)' = \cos x$$
$$f\big(g(x)\big)' = f'\big(g(x)\big)\, g'(x)$$

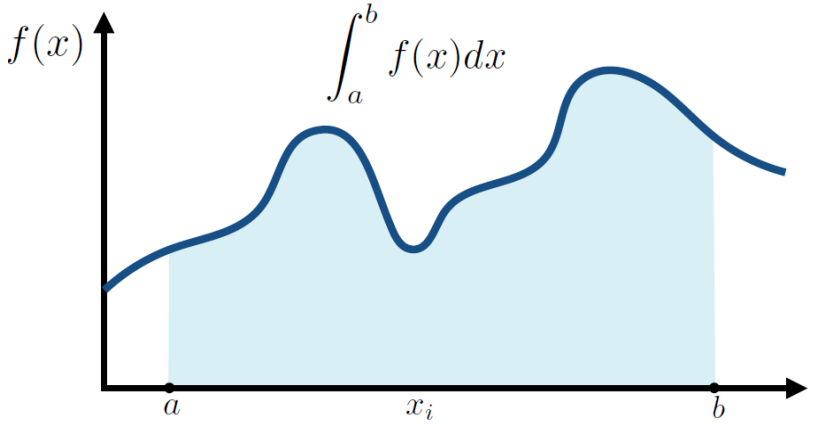
Part II
Solving the Rendering Equation

Pause

Solving the Rendering Equation

$$L_o(x, \omega_o) = L_e(x, \omega_o) + \int_{\Omega_{2\pi}} L_i(x, \omega_i)\, f_r(\omega_o, \omega_i)\, (N \cdot \omega_i)\, \mathrm{d}\omega_i$$

- The rendering equation has no analytic solution except for very cases where very strong assumptions are made
  - Infinitely recursive (the same quantity $L(x, \omega)$ appears on both sides of the equation)
  - The value of L_i can only be known through point sampling
  - Need to resort to numerical methods to provide an approximation
  - Most common: ***Monte Carlo integration***



Monte Carlo Integration – Problem Setup

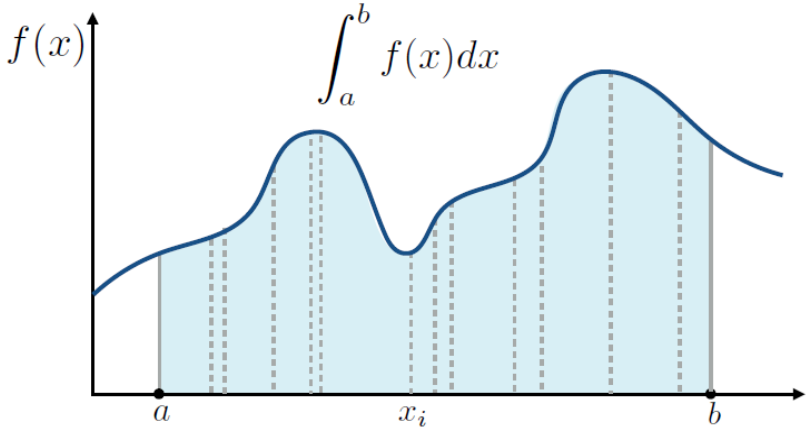
Goal:

Compute $I = \int_a^b f(x)\,\mathrm{d}x$

Typical constraint:

$f(x)$ can only be known through sampling

Monte Carlo Integration - Overview



Estimate the integral by ***randomly*** sampling the integrand

General formula: $\hat{I} = \frac{1}{N}\sum_{j=1}^{N}\frac{f(x_j)}{p(x_j)}$

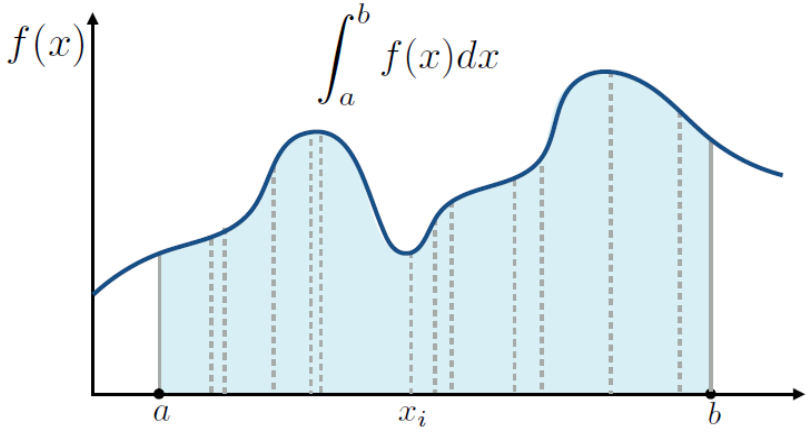
N is the number of used samples

x_j is the sample location

$f(x_j)$ is the sample value (or observation)

$p(x_j)$ is the probability of choosing x_j

Monte Carlo Integration - Overview



$$\hat{I} = \frac{1}{N}\sum_{j=1}^{N}\frac{f(x_j)}{p(x_j)}$$

Advantages:

- Simple (both conceptually and in practice)
- Dimension independent
- Works for complex functions

Disadvantages:

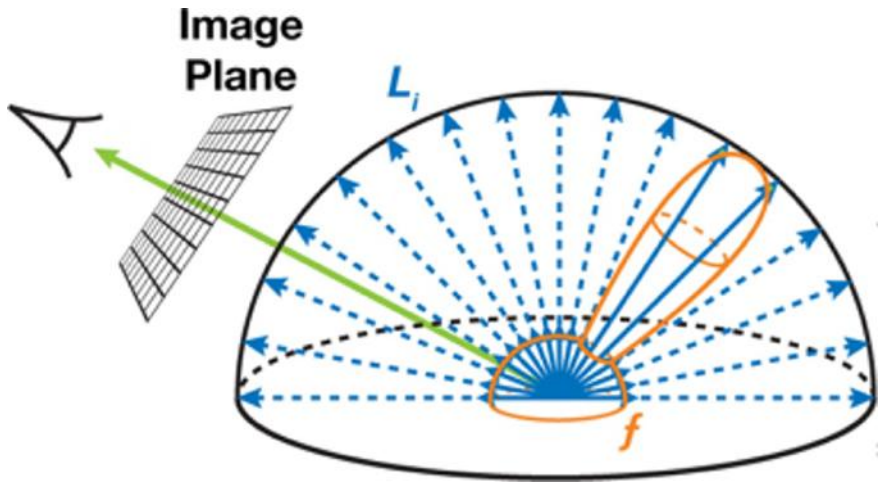
- Slow convergence (needs many samples)
- Variance (leads to noisy images)



Monte Carlo Integration Noise

Monte Carlo for the RE

$$L_o(x,\omega_o) \approx L_e(x,\omega_o) + \frac{1}{N}\sum_{j=1}^{N}\frac{L_i(x,\omega_j)\, f_r(\omega_o,\omega_j)\,(N\cdot\omega_j)}{p(\omega_j)}$$



Direct vs Indirect Illumination

- Direct Illumination: the light that arrives at a surface directly from a light source
  - Hard transitions, sharp variations
  - We know where it comes from (light sources)
- Indirect Illumination: all the light that arrives after bouncing at least once
  - Smooth, slow variations
  - Can come from any direction above the surface
- Different type of light with very different features → Convenient to treat them separately





Direct + Indirect = Final Image

Splitting the RE into Direct and Indirect

- Let us state that $L_i(x, \omega_i) = L_i^{dir}(x, \omega_i) + L_i^{ind}(x, \omega_i)$
- Then, we can re-write the rendering equation as follows:

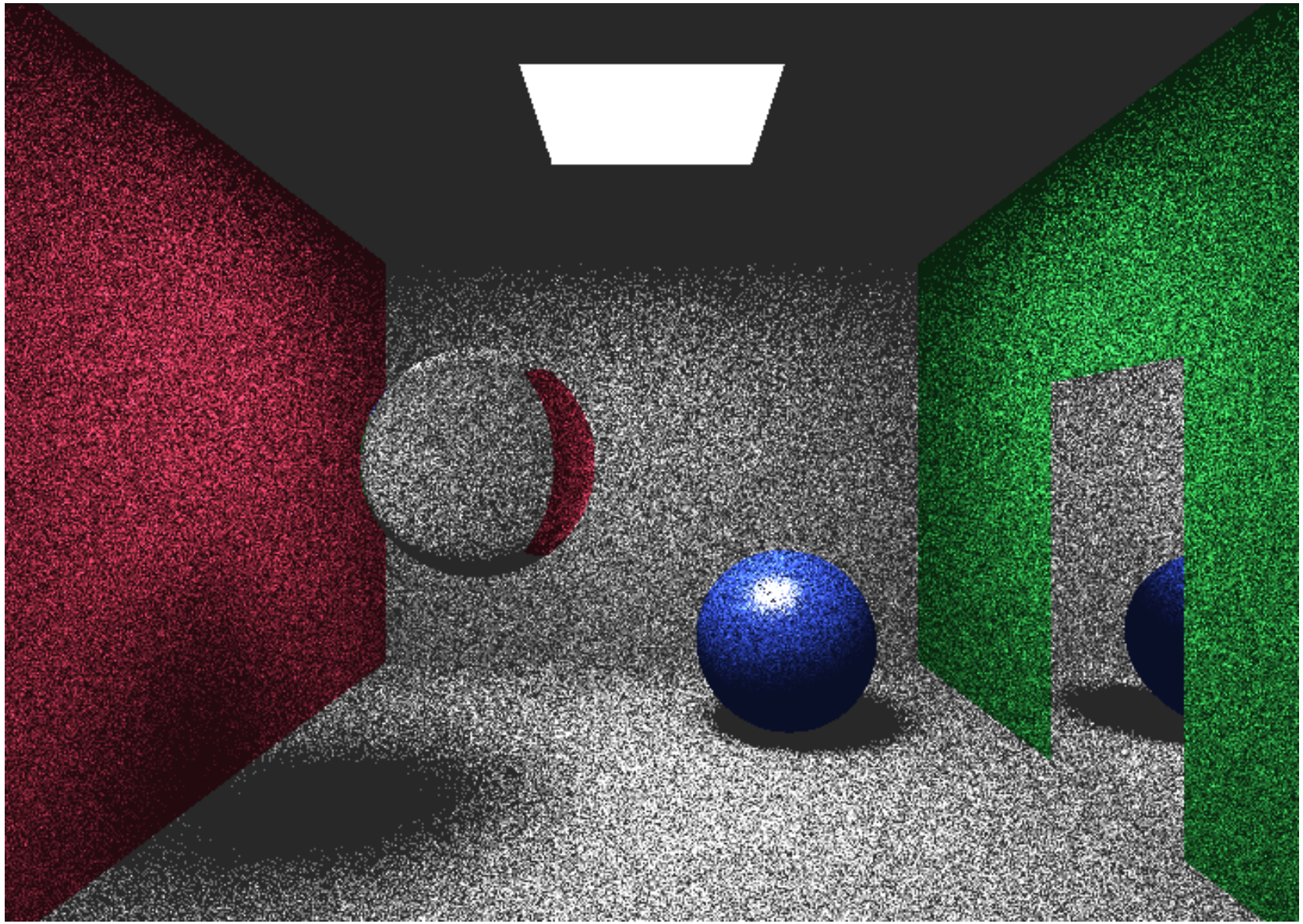
$$L_o(x, \omega_o) = L_e(x, \omega_o) + \underbrace{\int_{\Omega_{2\pi}} L_i^{dir}(x, \omega_i)\, f_r(\omega_o, \omega_i)\, (N \cdot \omega_i)\, \mathrm{d}\omega_i}_{L_o^{dir}(x, \omega_o)} + \underbrace{\int_{\Omega_{2\pi}} L_i^{ind}(x, \omega_i)\, f_r(\omega_o, \omega_i)\, (N \cdot \omega_i)\, \mathrm{d}\omega_i}_{L_o^{ind}(x, \omega_o)}$$

- Each integral can be estimated separately
- In the following, we'll delve into the direct illumination integral

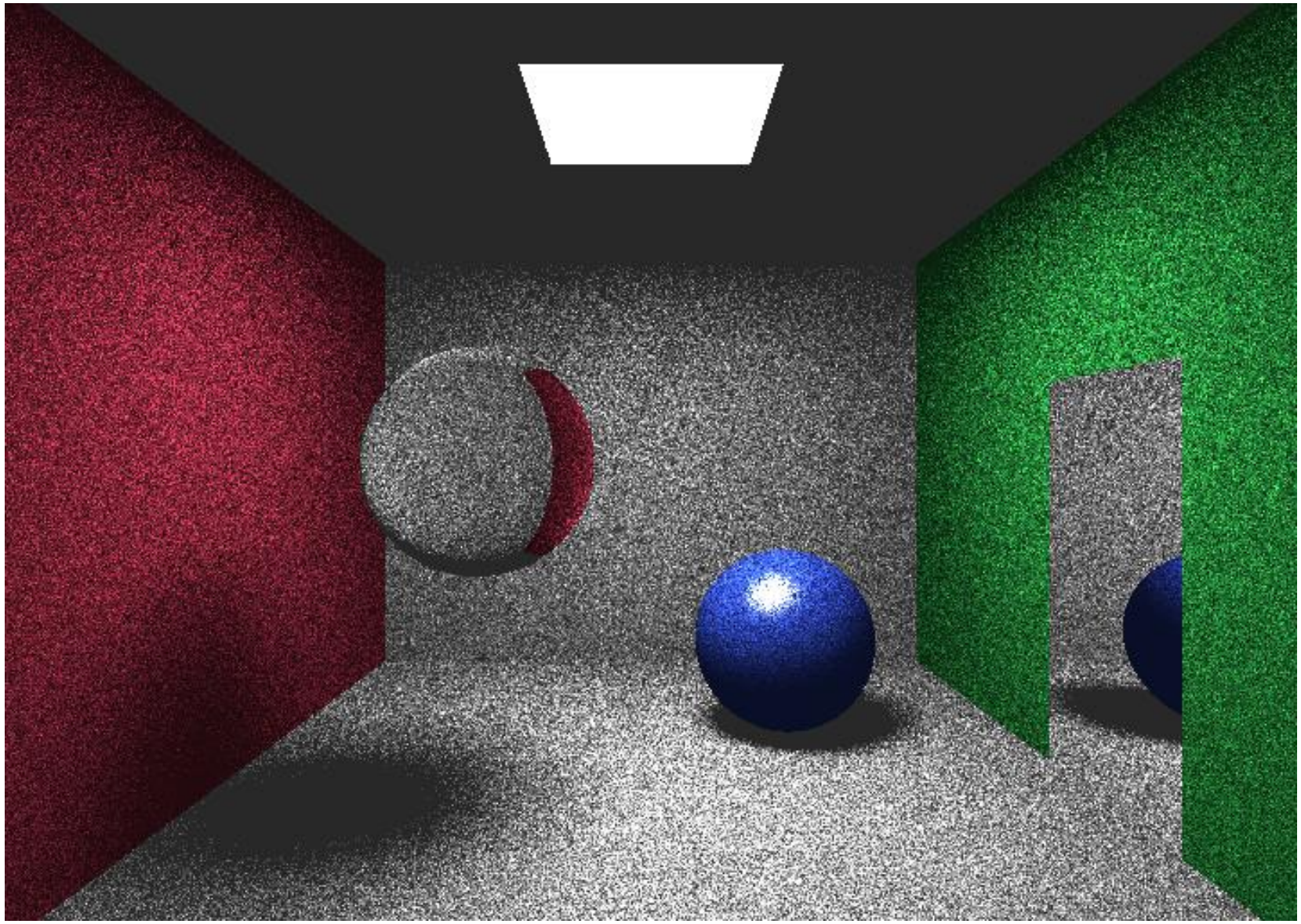
Estimating the Direct Illumination

- Simple strategy
  - At each visible point generate a set of uniformly distributed random directions over the hemisphere ($p(\omega) = 1/2\pi$)
  - If we touch a light source, then account for its contribution
  - If not, then $L_i^{dir}(x, \omega_j) = 0$

$$L_o^{dir}(x, \omega_o) = \frac{2\pi}{N} \sum_{j=1}^{N} L_i^{dir}(x, \omega_j)\, f_r(\omega_o, \omega_j)(N \cdot \omega_j)$$



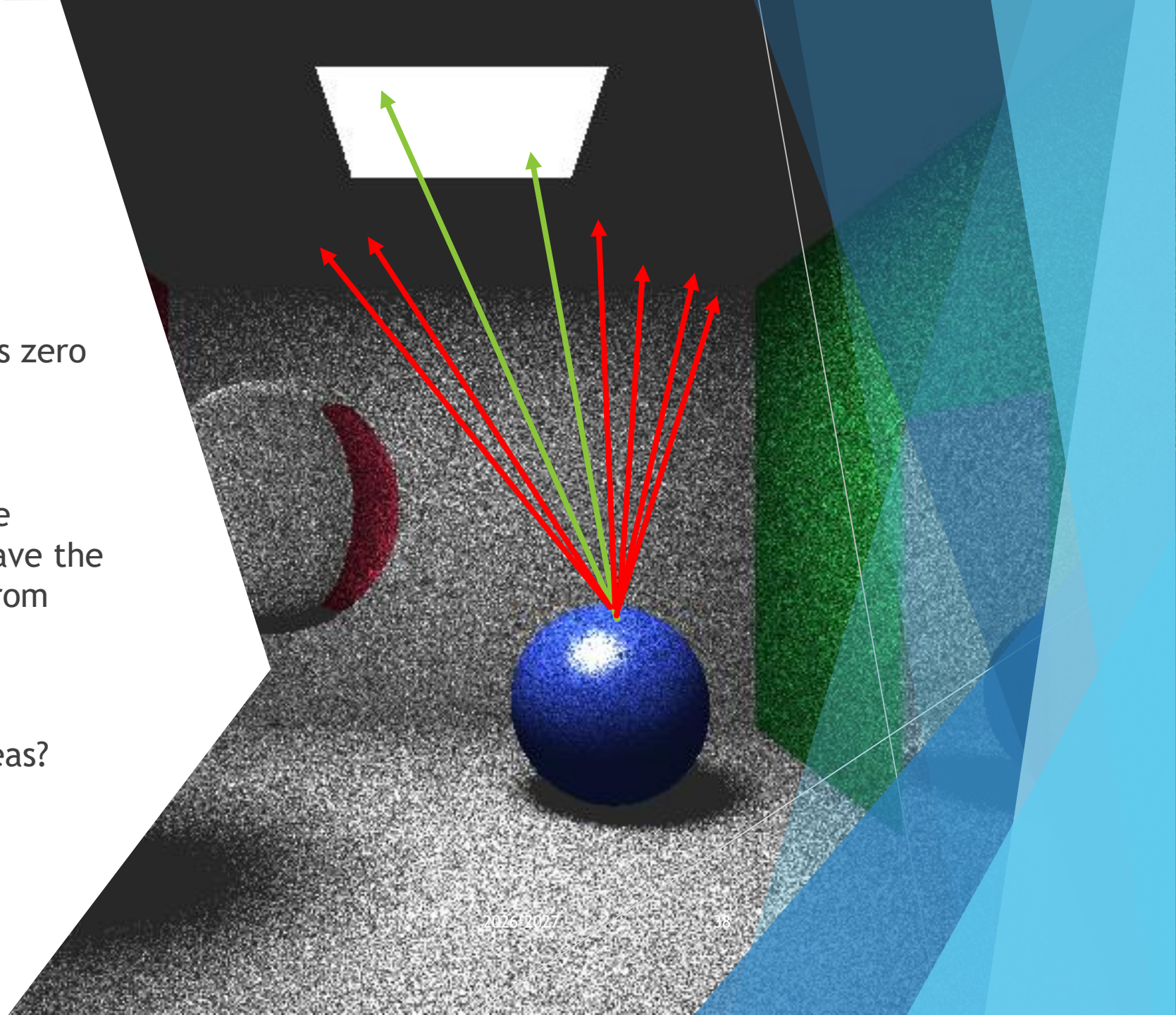
128 samples



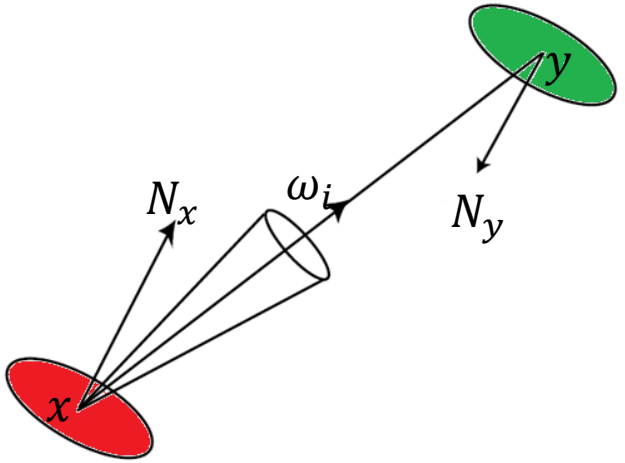
256 samples

Observation

- Incoming radiance $L_i^{dir}(x, \omega_j)$ is zero for most samples
- Only the samples that have the direction of the light source have the possibility of being different from zero (if there is no occlusion)
- How can we do better? Any ideas?



Area Formulation of the Direct Component



- Alternative formulation of the RE
  - Instead of integrating over all possible directions (i.e., solid angles ω_i)…
  - … integrate over all the surfaces of the scene represented by the set A
  - Mathematically, this is a change of integration variable: $d\omega_i = \frac{(-\omega_i)\cdot N_y}{\|x-y\|^2} dA$, yielding:

$$L_o^{dir}(x,\omega_o) = \int_A L_e(y,-\omega_i)\, f_r(\omega_o,\omega_i)\, G(x,y) V(x,y)\, \mathrm{d}A_y$$

where:

- $\omega_i = (y-x)/\|y-x\|$
- y is a differential surface element
- $G(x,y) = \frac{(N_x\cdot\omega_i)\left((-\omega_i)\cdot N_y\right)}{\|y-x\|^2}$ is the geometry term and depends on the relative geometry of the surfaces at point x and y

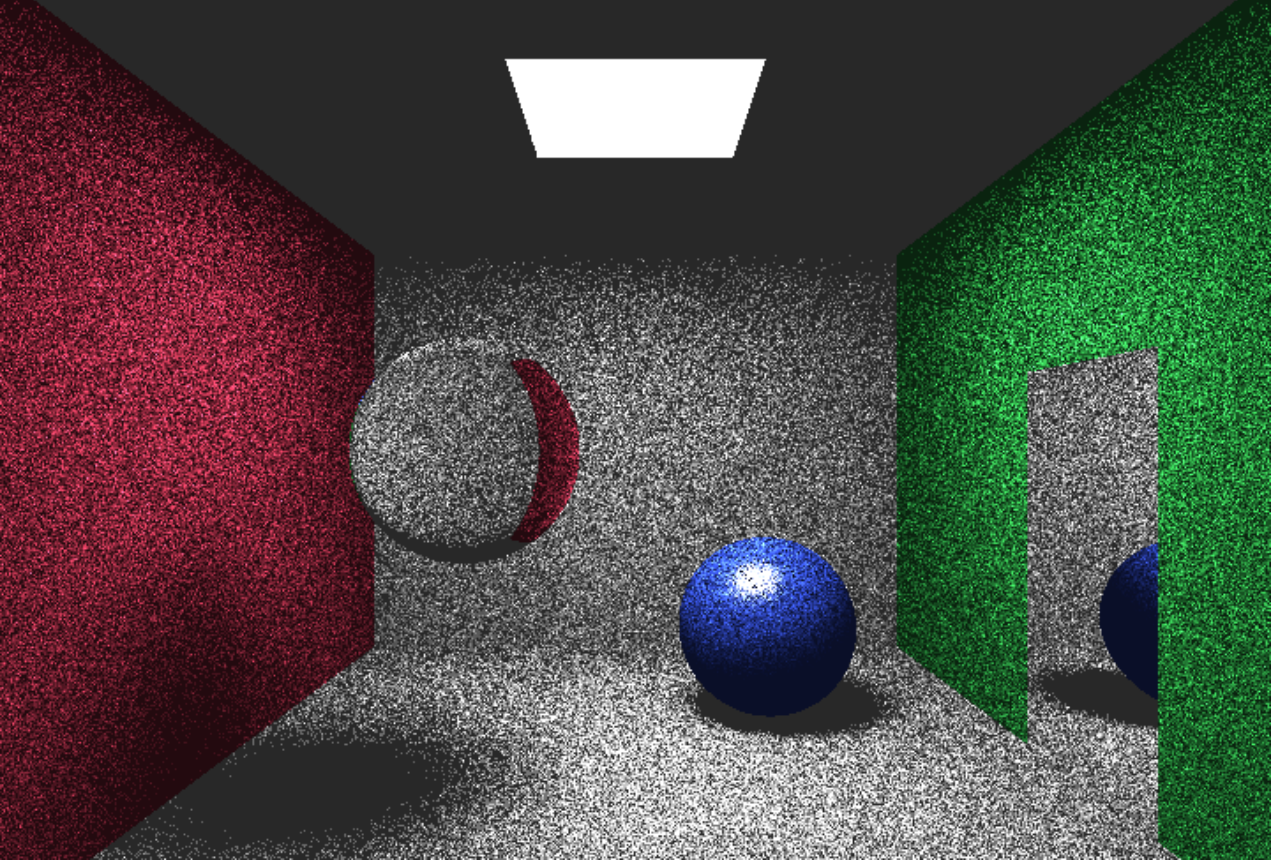
Estimating the Direct Illumination
Area Formulation

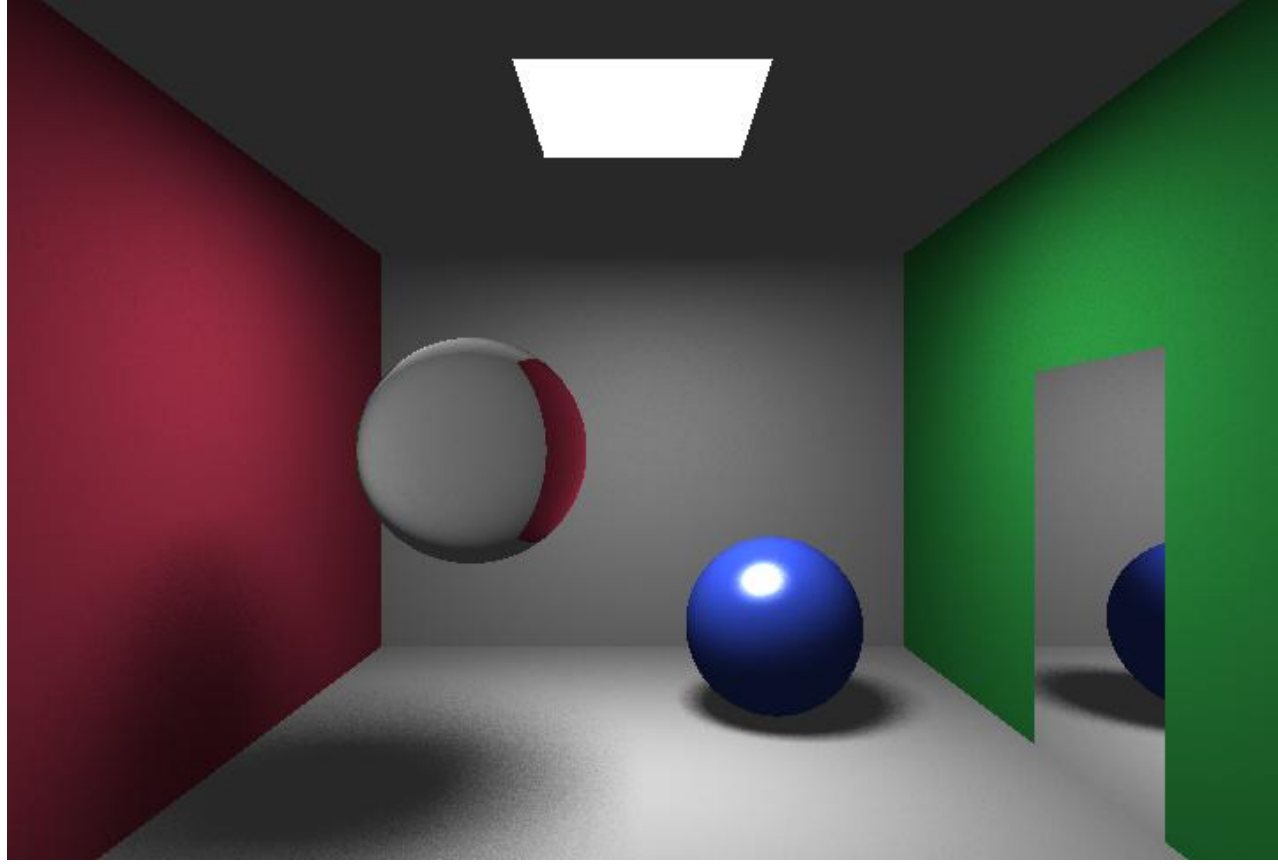
- Assume we have a single light source with area A_L
- Generate a set of samples y_j uniformly distributed on the light source such that $p(y_j) = 1/A_L$
- If the point y_j is visible from x then $V(x, y_j) = 1$ and y_j contributes to the illumination at x
- If not, then there is an occlusion and thus no light contribution

$$L_o^{dir}(x, \omega_o) \approx \frac{1}{N} \sum_{j=1}^{N} \frac{L_e(y_j, -\omega_i) f_r(\omega_o, \omega_i)\, G(x, y_j)\, V(x, y_j)}{p(y_j)}$$

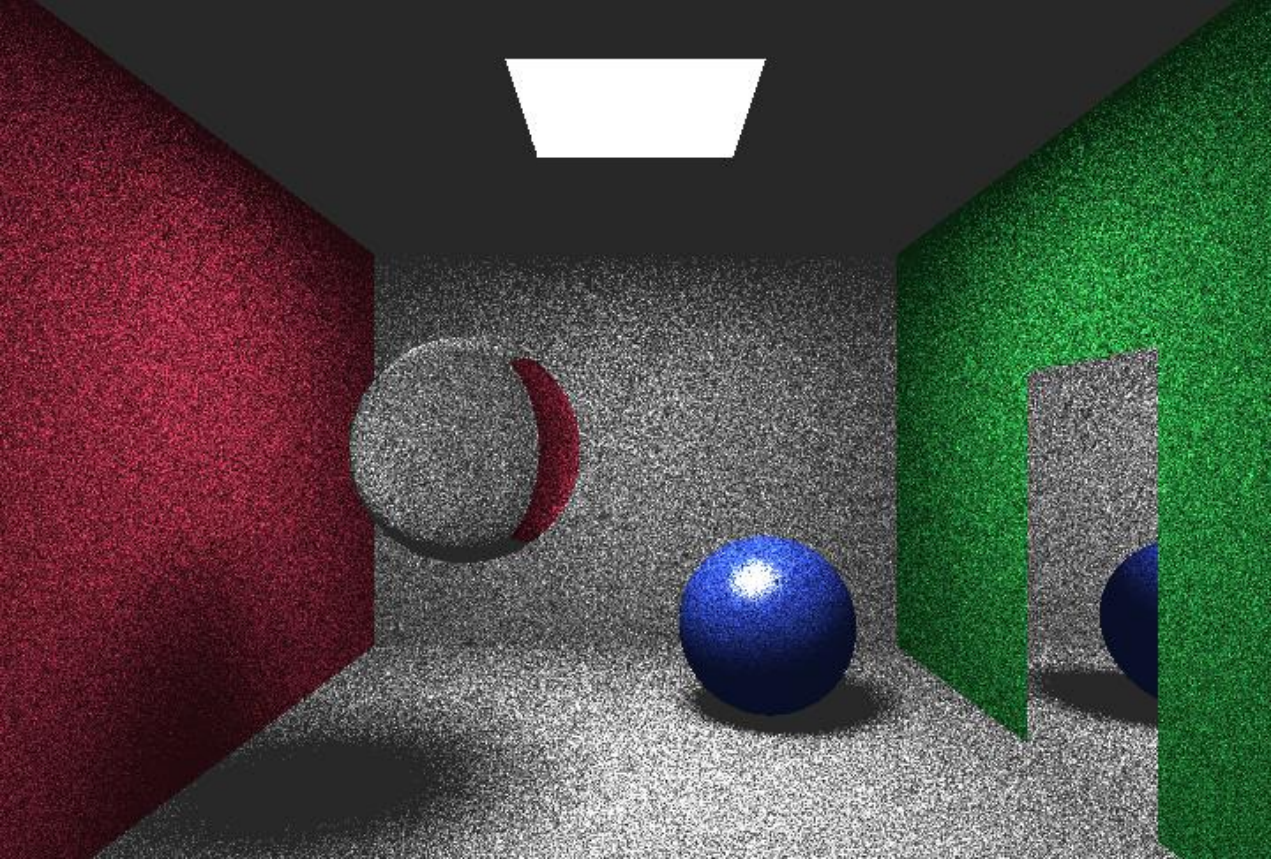
$$= \frac{A_L}{N} \sum_{j=1}^{N} L_e(y_j, -\omega_i) f_r(\omega_o, \omega_i)\, G(x, y_j)\, V(x, y_j)$$

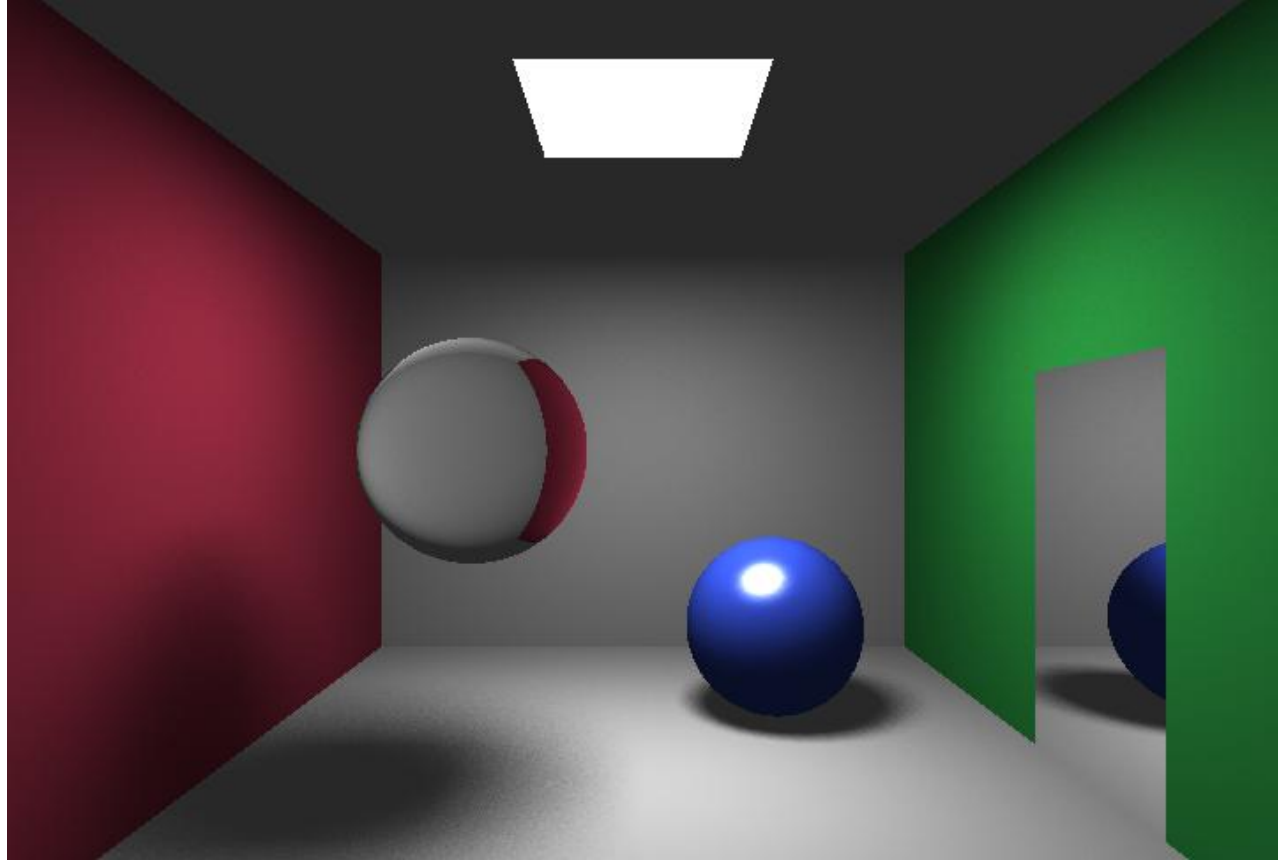
where $\omega_i = \frac{y_i - x}{\| y_i - x \|}$





Left: Solid Angle Sampling (128 samples)
Right: Area Sampling (128 samples)





Left: Solid Angle Sampling (256 samples)
Right: Area Sampling (256 samples)

Conclusion

In a Nutshell

- Radiance is the physical quantity which captures the appearance of real-world objects
- Physically-Based Rendering is all about computing radiance
- Radiance is computed using the Rendering Equation
  - No analytic solution (in general)
  - Common to resort to numerical methods (e.g., Monte Carlo integration)
- Monte Carlo methods are costly, and need many samples to converge
  - However, we can accelerate convergence by trying to sample the most relevant light paths

Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Credits

- Thank you to Josep Blat for making available the slides for ACG (2023)
- Several slides and images are inspired on Berkeley's CS184/284A course on Computer Graphics
- Several slides and images are inspired on the slides of Kadi Bouatouch on Radiometry and on Stochastic Ray Tracing



The End

Questions?